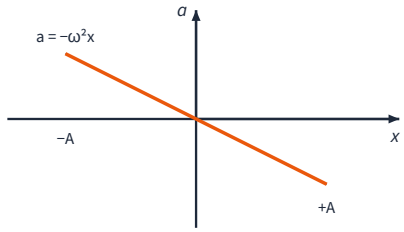
CHAPTER 20

OSCILLATIONS

Key idea: in SHM the acceleration is proportional to the displacement and directed towards equilibrium.

1 SIMPLE HARMONIC MOTION



$$a = -\omega^2 x \qquad F = -kx$$

- $a \propto x$, always towards the equilibrium point
- $\omega = 2\pi f = \dfrac{2\pi}{T}$
- a–x graph: straight line, gradient $-\omega^2$

2 KINEMATICS OF SHM

$$x = A\sin\omega t$$

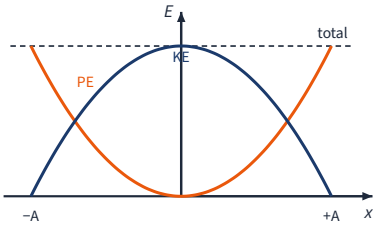
$$v = \omega A\cos\omega t$$

$$v = \pm\omega\sqrt{A^2 - x^2}$$

$$v_{\max} = \omega A \qquad a_{\max} = \omega^2 A$$

$x = A\cos\omega t$ if timing starts at an extreme.

3 ENERGY IN SHM



$$E = \tfrac{1}{2}m\omega^2 A^2 = \tfrac{1}{2}kA^2$$

$$E_k = \tfrac{1}{2}m\omega^2(A^2 - x^2)$$

$$E_p = \tfrac{1}{2}m\omega^2 x^2$$

Total energy $\propto$ (amplitude)^2; $E_k = E_p$ at $x = A/\sqrt{2}$.

4 SYSTEMS IN SHM

MASS-SPRING

$$T = 2\pi\sqrt{\frac{m}{k}} \qquad \omega = \sqrt{\frac{k}{m}}$$

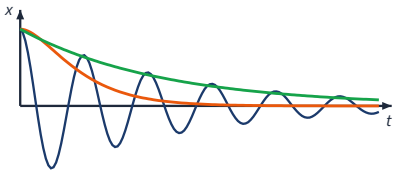
SIMPLE PENDULUM (SMALL Θ)

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Independent of mass and amplitude; $\theta \lesssim 10^\circ$.

Springs: parallel $k = k_1 + k_2$; series $\dfrac{1}{k} = \dfrac{1}{k_1} + \dfrac{1}{k_2}$

5 DAMPED OSCILLATION

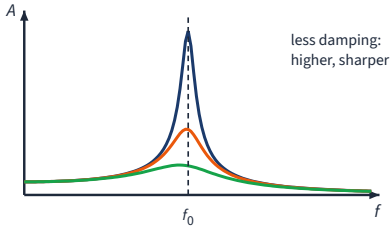


light (navy), critical (orange), heavy (green)

Light	oscillates, amplitude decays exponentially
Critical	back to equilibrium fastest, no oscillation
Heavy	slow return, no oscillation

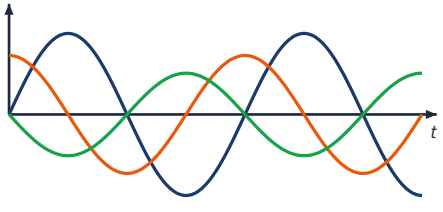
$$A = A_0 e^{-bt}$$

6 FORCED OSCILLATION & RESONANCE



- Forced: oscillates at the **driving** frequency
- Resonance: $f_{\text{driver}} = f_0$, maximum amplitude
- More damping: lower, broader peak, shifted slightly to lower f

7 PHASE RELATIONSHIPS



x (navy), v (orange), a (green)

v leads x by $\pi/2$; a is in antiphase (π) with x.

8 COMMON SECTION A TRAPS

- Calculator in degrees for $\sin\omega t$: use **radians**.
- Saying the period depends on amplitude (it does not).
- Pendulum length: to the **centre** of the bob.
- Doubling A doubles $v_{\max}$ but **quadruples** E.
- Speed is maximum at $x = 0$, acceleration at $x = \pm A$.

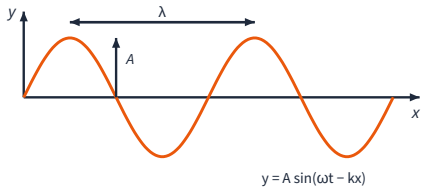
EXAM TIPS

- ✓Prove SHM by showing $a = -(\text{constant})x$.
- ✓Identify ω as the coefficient of t.
- ✓Use energy for speed at a given x.
- ✓Sketch x, v, a graphs one under another, $T/4$ apart.

CHAPTER 21 WAVE MOTION

Key idea: a progressive wave carries energy without carrying matter; overlapping waves add.

1 PROGRESSIVE WAVES



$$v = f\lambda$$

$$y = A\sin(\omega t - kx)$$

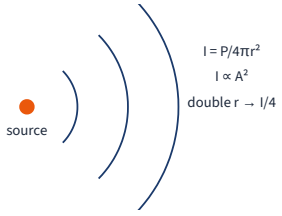
- $k = \dfrac{2\pi}{\lambda}, v = \dfrac{\omega}{k}$
- $+kx$: travels in the $-x$ direction
- Transverse: $\perp$ to travel; longitudinal: $\parallel$

2 PHASE DIFFERENCE

$$\phi = \frac{2\pi}{\lambda}\Delta x \qquad \phi = \frac{2\pi}{T}\Delta t$$

In phase: $\Delta x = n\lambda$. Antiphase: $\Delta x = (n + \frac{1}{2})\lambda$.

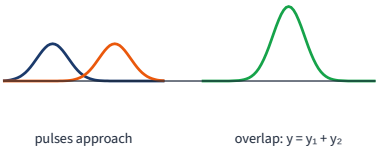
3 INTENSITY



$$I = \frac{P}{4\pi r^2} \qquad I \propto A^2$$

Power per unit area, W m^{-2}. Point source: $I \propto 1/r^2$ so $A \propto 1/r$.

4 SUPERPOSITION



Constructive path difference $n\lambda$

Destructive path difference $(n + \frac{1}{2})\lambda$

Coherent: same frequency, constant phase difference.

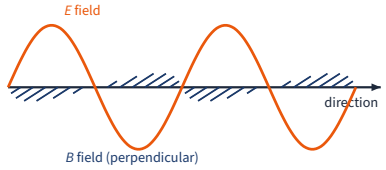
5 STANDING WAVES

$$y = 2A\sin kx\ \cos\omega t$$

- Two equal waves travelling in **opposite** directions
- Node to node $= \lambda/2$; node to antinode $= \lambda/4$
- String fixed both ends: $L = \dfrac{n\lambda}{2}$, $f_n = \dfrac{nv}{2L}$

$$v = \sqrt{\frac{T}{\mu}}$$

6 ELECTROMAGNETIC WAVES



$$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \qquad E_0 = cB_0$$

$E \perp B \perp$ direction, in phase. Radio → microwave → IR → visible → UV → X-ray → γ (λ decreasing).

7 PROGRESSIVE VS STANDING

	Progressive	Standing
Energy	transferred	not transferred
Amplitude	same everywhere	zero at nodes, max at antinodes
Phase	changes along the wave	same within a loop; antiphase in next loop
Wave profile	moves	does not move

8 COMMON SECTION A TRAPS

- Node spacing is $\lambda/2$, not λ.
- Reading k as the wavelength; $\lambda = 2\pi/k$.
- Doubling the distance quarters I but halves A.
- μ is mass per unit **length** (kg m^{-1}).

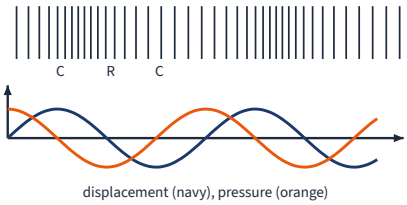
EXAM TIPS

- ✓From $y = A\sin(\omega t - kx)$ read A, ω and k first.
- ✓Draw the harmonics with N and A labelled.
- ✓Use ratios for the inverse square law.
- ✓State the sources are coherent before discussing interference.

CHAPTER 22 SOUND WAVES

Key idea: sound is a longitudinal pressure wave; air columns and strings resonate at definite frequencies.

1 PROPAGATION OF SOUND

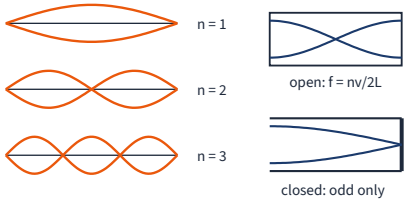


displacement (navy), pressure (orange)

$$v = \sqrt{\frac{\gamma RT}{M}} \qquad v = \sqrt{\frac{E}{\rho}}$$

- Longitudinal: compressions and rarefactions
- Needs a medium; faster in solids
- Gas: $v \propto \sqrt{T}$, independent of pressure

2 AIR COLUMNS



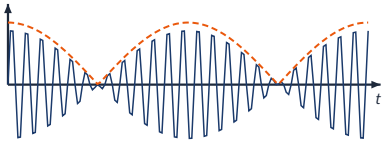
Pipe	Ends	Harmonics	f_1
Open	A–A	all	$v/2L$
Closed	N–A	odd only	$v/4L$

3 INTENSITY LEVEL

$$\beta = 10 \log_{10} \frac{I}{I_0}$$

- $I_0 = 1.0 \times 10^{-12}$ W m^{-2} (threshold of hearing)
- +10 dB = intensity ×10; +3 dB ≈ ×2
- N identical sources: $\beta + 10 \log_{10} N$
- Add intensities, **not** decibels

4 BEATS



loudness rises and falls $f_1 - f_2$ times per second

$$f_{\text{beat}} = |f_1 - f_2|$$

$$f = \frac{f_1 + f_2}{2}$$

Loudness rises and falls as the waves move in and out of phase.

5 DOPPLER EFFECT

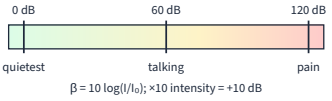
$$f' = f \frac{v \pm v_o}{v \mp v_s}$$

- Upper signs: **approaching** (frequency rises)
- Speeds relative to the air
- Moving reflector: apply twice (observer, then source)

6 STRINGS & END CORRECTION

$$f_n = \frac{n}{2L} \sqrt{\frac{T}{\mu}}$$

- Node at each fixed end; all harmonics
- Closed pipe: $L + e = \frac{\lambda}{4}$
- Antinode lies slightly **beyond** the open end



β = 10 log(I/I₀); ×10 intensity = +10 dB

7 UNIT CHECK

Quantity	Formula	Unit
Intensity	$P/4\pi r^2$	W m^{-2}
Intensity level	$10 \log(I/I_0)$	dB
Beat frequency	$\lvert f_1 - f_2 \rvert$	Hz
Mass per length	μ	kg m^{-1}

8 COMMON SECTION A TRAPS

- Closed pipe has **odd** harmonics only: 1st, 3rd, 5th…
- Adding dB values directly.
- Using °C in $v \propto \sqrt{T}$.
- Doppler: the wrong sign for approach.
- Displacement node = pressure **antinode**.

EXAM TIPS

- ✓Sketch the air column with N and A before using a formula.
- ✓Convert dB to intensity, add, then convert back.
- ✓Write f' formula with signs explained in words.
- ✓Use $v = 4f(L + e)$ for resonance-tube data.

CHAPTER 23

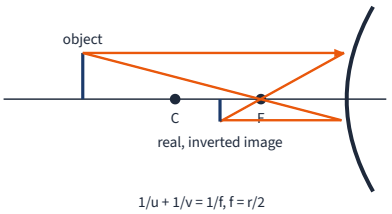
GEOMETRICAL OPTICS

Key idea: mirrors and lenses form images by reflection and refraction; use the real-is-positive convention.

1 SIGN CONVENTION

- Real object, real image: **positive**
- Virtual image: v **negative**
- Concave mirror, convex lens: $f > 0$
- Convex mirror, concave lens: $f < 0$
- $m = -\dfrac{v}{u}$: negative = inverted, positive = upright

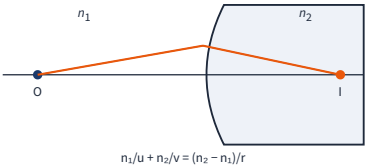
2 SPHERICAL MIRRORS



1/u + 1/v = 1/f, f = r/2

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \qquad f=\frac{R}{2}$$

3 REFRACTION AT A CURVED SURFACE



$n_1/u + n_2/v = (n_2 - n_1)/r$

$$\frac{n_1}{u}+\frac{n_2}{v}=\frac{n_2-n_1}{R}$$

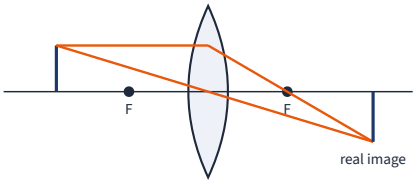
- $R > 0$ if the centre is on the **outgoing** side
- Plane surface: apparent depth $= \dfrac{\text{real depth}}{n}$

4 IMAGE NATURE

Object (converging)	Image
$u > 2f$	real, inverted, diminished
$u = 2f$	real, inverted, same size
$f < u < 2f$	real, inverted, magnified
$u < f$	virtual, upright, magnified

Convex mirror / concave lens: always virtual, upright, diminished.

5 THIN LENSES



1/u + 1/v = 1/f

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$$

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \qquad P=\frac{1}{f}$$

Biconvex: both R positive. Power in dioptre (D), f in metres.

6 USEFUL RESULTS

- Lenses in contact: $\dfrac{1}{f}=\dfrac{1}{f_1}+\dfrac{1}{f_2}$, $P = P_1 + P_2$
- Same-size real image: $u = v = 2f$
- Minimum object–image distance: $4f$
- Mirror: f does not depend on the medium; lens: f does

7 RAY-DIAGRAM RULES

1. Ray parallel to the axis → through (or from) F.
2. Ray through the optical centre (lens) or C (mirror) → undeviated / back on itself.
3. Ray through F → emerges parallel to the axis.

8 COMMON SECTION A TRAPS

- Dropping the minus sign of a virtual image or a diverging lens.
- Using $f = R$ instead of $R/2$ for a mirror.
- Wrong sign of R in the lensmaker's or surface equation.
- Mixing cm and m when finding power in D.

EXAM TIPS

- ✓State the sign convention at the start of the answer.
- ✓Draw a quick ray diagram to check the answer is sensible.
- ✓Use $m = -v/u$ and describe the image in three words.
- ✓Find each image in turn for two-lens systems.

CHAPTER 24 WAVE OPTICS

Key idea: light shows interference, diffraction and polarisation, so it is a transverse wave.

1 HUYGENS'S PRINCIPLE



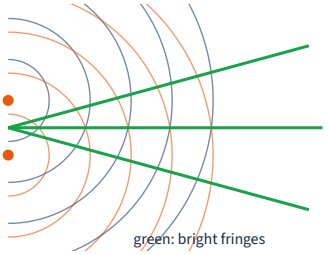
each point is a source of secondary wavelets; new wavefront = envelope

Every point on a wavefront is a source of secondary wavelets; the new wavefront is their **envelope**.

$$n = \frac{c}{v} = \frac{\lambda_0}{\lambda}$$

On refraction f is unchanged; v and λ change.

2 YOUNG'S DOUBLE SLIT

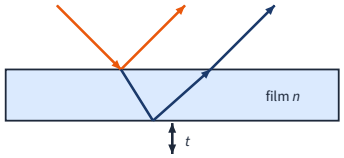


green: bright fringes

$$x = \frac{\lambda D}{a} \qquad a \sin\theta = n\lambda$$

- Bright: path difference $n\lambda$; dark: $(n + \frac{1}{2})\lambda$
- White light: white centre, coloured fringes (violet nearest)

3 THIN FILMS



path difference 2nt (+ λ/2 for one reflection phase change)

- π phase change on reflection at a **denser** medium
- Film in air (one change): bright $2nt = (m + \frac{1}{2})\lambda$
- Coating on glass (two changes): dark $2nt = (m + \frac{1}{2})\lambda$
- Minimum coating: $t = \dfrac{\lambda}{4n}$
- Air wedge: $x = \dfrac{\lambda}{2\theta}$, dark at contact

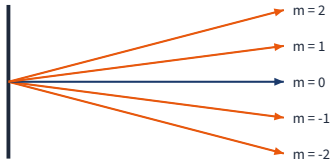
4 SINGLE-SLIT DIFFRACTION



$$a \sin\theta = m\lambda \qquad w = \frac{2\lambda D}{a}$$

Minima ($m = 1, 2, \ldots$). Central maximum twice as wide as the others.

5 DIFFRACTION GRATING

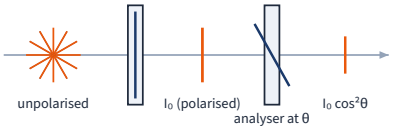


d sin θ = mλ: sharp, bright maxima

$$d \sin\theta = m\lambda \qquad d = \frac{1}{N}$$

- Highest order: largest integer below d/λ
- More lines: sharper, brighter maxima
- Total maxima $= 2m_{\max} + 1$

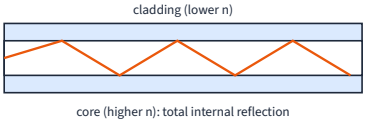
6 POLARISATION



$$I = I_0 \cos^2\theta \qquad \tan\theta_B = n$$

- Unpolarised through a polariser: $I_0/2$
- At θ_B: reflected and refracted rays at 90°
- Only **transverse** waves can be polarised

7 OPTICAL FIBRES



$$\sin\theta_c = \frac{n_2}{n_1} \qquad \mathrm{NA} = \sqrt{n_1^2 - n_2^2}$$

Total internal reflection: $n_{\text{core}} > n_{\text{cladding}}$, angle > θ_c. Modal dispersion spreads pulses; attenuation $= \dfrac{10}{L} \log \dfrac{P_{\text{in}}}{P_{\text{out}}}$ dB per unit length.

8 COMMON SECTION A TRAPS

- Mixing a (slit separation or width) with d (grating spacing).
- Forgetting the extra λ/2 from a reflection at a denser medium.
- Lines per mm → lines per m before $d = 1/N$.
- Using Malus's law on **unpolarised** light.
- Single slit: $a \sin\theta = m\lambda$ gives **minima**.

EXAM TIPS

- ✓State "coherent sources" and the path-difference condition.
- ✓For thin films, count the phase changes first.
- ✓Check $\sin\theta \leq 1$ for the highest grating order.
- ✓Fringe spacing ∝ λ: red fringes are wider than blue.

CHAPTER 25

QUANTUM PHYSICS

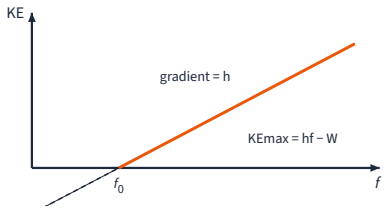
Key idea: light comes in photons of energy hf, and moving particles behave as waves of wavelength h/p.

1 PHOTON

$$E = hf = \frac{hc}{\lambda} \qquad p = \frac{h}{\lambda}$$

- A quantum (packet) of electromagnetic energy
- 1 eV $= 1.60 \times 10^{-19}$ J
- E (eV) ≈ 1240 ÷ λ (nm)

2 PHOTOELECTRIC EFFECT

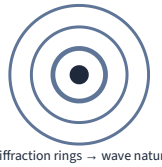


$$hf = \phi + K_{\max}$$

$$eV_s = K_{\max}$$

Graph of $K_{\max}$ against f: gradient h, f-intercept $f_0 = \phi/h$.

3 WAVE-PARTICLE DUALITY



electron diffraction rings → wave nature, λ = h/p

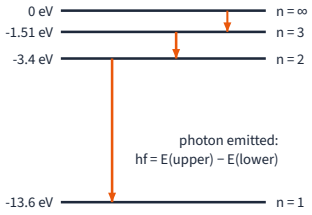
$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2meV}}$$

Electron diffraction rings show the wave nature of electrons.

4 WAVE THEORY FAILS

1. There is a **threshold frequency** f_0.
2. $K_{\max}$ depends on frequency, **not** intensity.
3. Emission is **instantaneous**.
4. Intensity changes only the number of photoelectrons.

5 ATOMIC STRUCTURE

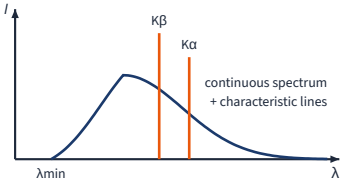


$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

$$hf = E_i - E_f$$

Lyman → $n = 1$ (UV); Balmer → $n = 2$ (visible); Paschen → $n = 3$ (IR). Bohr: $mvr = \frac{nh}{2\pi}$.

6 X-RAYS



$$\lambda_{\min} = \frac{hc}{eV} \qquad 2d\sin\theta = n\lambda$$

- Continuous spectrum: braking (bremsstrahlung)
- Sharp lines: characteristic of the target
- Bragg angle θ is measured from the **planes**

7 NANOSCIENCE

- Size 1–100 nm; huge surface-area-to-volume ratio ($3/r$ for a sphere)
- Large fraction of atoms at the surface → high reactivity and catalytic activity
- STM: tunnelling current depends exponentially on the gap

8 COMMON SECTION A TRAPS

- Using eV in $hf = \phi + K$ while h is in J s.
- Saying brighter light gives faster photoelectrons.
- Forgetting energy levels are **negative**.
- Using the electron mass for a proton in $\lambda = h/p$.
- $\lambda_{\min}$ depends on the voltage, not the target.

EXAM TIPS

- ✓Convert every energy to joules (or every one to eV) before substituting.
- ✓Draw the transition arrow on the level diagram.
- ✓For $\lambda_{\min}$, all the electron KE becomes one photon.
- ✓Quote $h = 6.63 \times 10^{-34}$ J s.

CHAPTER 26 NUCLEAR PHYSICS

Key idea: the mass defect of a nucleus is its binding energy; unstable nuclei decay randomly.

1 THE NUCLEUS

$${}^{A}_{Z}\mathrm{X}$$

- Z: protons; A: nucleons; $N = A - Z$
- Isotopes: same Z, different A
- $R = R_0 A^{1/3}$, $R_0 \approx 1.2$ fm
- Nuclear density the same for all nuclei

2 MASS DEFECT & BINDING ENERGY

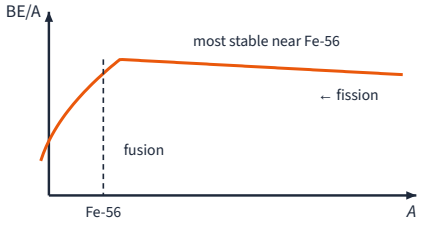
$$\Delta m = Z m_p + (A - Z) m_n - m_{\text{nucleus}}$$

$$B = \Delta m\, c^2$$

$$1\ \text{u} = 931.5\ \text{MeV}$$

Binding energy: energy needed to separate a nucleus into its nucleons.

3 BINDING ENERGY PER NUCLEON



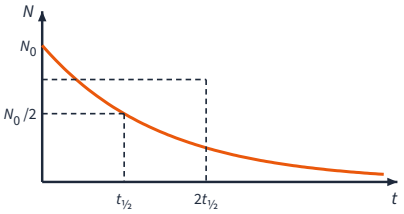
- Peak ≈ 8.8 MeV near ^{56}Fe: most stable
- Light nuclei → **fusion**; heavy nuclei → **fission**

4 RADIOACTIVE DECAY

Emission	Change
α ($^4_2$He)	$A - 4$, $Z - 2$
β^-	A same, $Z + 1$
γ	no change

- **Spontaneous**: not affected by conditions
- **Random**: cannot predict which nucleus decays next

5 DECAY LAW



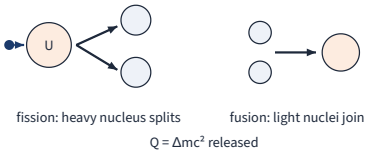
$$N = N_0 e^{-\lambda t} \qquad A = \lambda N$$

$$t_{1/2} = \frac{\ln 2}{\lambda}$$

$$N = N_0 \left(\tfrac{1}{2}\right)^{t/t_{1/2}}$$

$\ln A$ against t: straight line, gradient $-\lambda$. Activity in Bq.

6 NUCLEAR REACTIONS



$$Q = \left(\sum m_{\text{reactants}} - \sum m_{\text{products}}\right) c^2$$

Conserved: A, Z (charge), energy, momentum. $Q > 0$: energy released.

7 FISSION VS FUSION

	Fission	Fusion
Process	heavy nucleus splits (e.g. ^{235}U + n)	light nuclei join (e.g. ^{2}H + ^{3}H)
Needs	slow neutron; chain reaction	very high temperature
Energy	≈ 200 MeV per fission	more per kg of fuel
Why	products have higher B/A	products have higher B/A

8 COMMON SECTION A TRAPS

- Using total binding energy instead of **per nucleon** for stability.
- Forgetting electron masses when using atomic masses.
- Taking $t_{1/2} = \lambda \ln 2$ (it is $\ln 2 / \lambda$).
- Mixing u, kg and MeV in one line.
- Time in years with λ in s^{-1}.

EXAM TIPS

- ✓Balance A and Z before finding Q.
- ✓Use 1 u = 931.5 MeV to skip c^2.
- ✓Count half-lives when t is a whole multiple of $t_{1/2}$.
- ✓Answer "why energy is released" with binding energy per nucleon.